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Charter

Identify capabilities and areas of potential instability of the "rest api todo list manager".
Identify documented and undocumented "rest api todo list manager" capabilities.
For each capability, create a script or small program to demonstrate the capability.
Exercise each capability identified with data typical to the intended use of the application.
Area: /todos/:id/tasksof(/:id) and /todos/:id/categories(/:id) (45 min)

Session setup

- runTodoManagerRestAPI-1.5.5.jar (v3), started fresh on port 4568
- Windows 10, Java 21, curl in Git Bash, docs at /docs

Summary of session findings

- link / list / unlink work for both relationships with {"id":"..."}, link returns 201 with empty body
- relationship GET/HEAD on a missing todo give 200 with other todos' data instead of 404 (linking from a missing todo does give 404)
- linking with an XML body does not work (404)
- posting a title instead of an id creates a new category (not documented)
- PUT with only a title drops all of the todo's links
- unlink twice -> 404, OPTIONS on a link only allows DELETE

Summary of concerns

- can't tell a todo is missing from the relationship endpoints
- XML users can't link categories
- PUT drops links as well as fields
- typo in a link body (title instead of id) quietly creates a category

Test ideas

- documented vs actual tests: missing todo 404, XML link body, PUT keeps links
- check the other side of each link (/projects/:id/tasks, /categories/:id/todos) - coordinate with P2/P3
- explore next: duplicate links, numeric id in link body, deleting a linked todo

Bugs logged

- GET /todos/:id/tasksof on a missing todo returns 200 with other todos' projects
- GET/HEAD /todos/:id/categories on a missing todo returns 200 instead of 404
- POST /todos/:id/categories with an XML body returns 404 for an existing category
- PUT /todos/:id with only a title removes the todo's category and project links

Decisions made during the session

- only the /todos/:id/... relationship endpoints, other side of links left to projects/categories sessions
- fresh app instance; created todo 3 to link (S2-05), seed todo 1 only read
- repeat-unlink only tried on categories (S2-13)
- linked todo 3 to a project and a category before the PUT to see if links survive (S2-19 to S2-21)
- saved request/response logs instead of screenshots

Areas not covered

- other side of each link (/projects/:id/tasks, /categories/:id/todos)
- duplicate links, numeric ids in link body
- deleting a todo that has links
- PUT/PATCH/OPTIONS on /todos/:id/tasksof

Files used or created

- session2_log.txt - all requests and responses (S2-01 to S2-24)
- session2_todo_relationships_capabilities.sh + session2_todo_relationships_capabilities_output.txt - script for each capability and its output on a fresh app
- runTodoManagerRestAPI-1.5.5.jar

Method	Method Type	is documented ?	Expected Behavior	expected == observed	Observed Behavior	Side Notes	XML & Screenshots
/todos/:id/tasksof	GET	DOCUMENTED	projects of todo	YES	200, project 1 (seed todo 1, and todo 3 after linking)		session2_log.txt S2-01, S2-07
/todos/:id/tasksof	HEAD	DOCUMENTED	headers only	YES	200, no body		session2_log.txt S2-02
/todos/:id/tasksof	POST	DOCUMENTED	link to project {"id"}	YES	201, empty body, shows up in GET		session2_log.txt S2-06/07
/todos/:id/tasksof	GET	DOCUMENTED	missing todo -> 404	NO	200, project 1 listed twice	#BUG	session2_log.txt S2-09
/todos/:id/tasksof/:id	DELETE	DOCUMENTED	unlink	YES	200		session2_log.txt S2-08
/todos/:id/categories	GET	DOCUMENTED	categories of todo	YES	200, category 1 for todo 1, category 2 for todo 3 after linking		session2_log.txt S2-03, S2-11
/todos/:id/categories	HEAD	DOCUMENTED	headers only	YES	200, no body		session2_log.txt S2-04
/todos/:id/categories	POST	DOCUMENTED	link to category {"id"}	YES	201, shows up in GET		session2_log.txt S2-10/11
/todos/:id/categories	POST	DOCUMENTED	link from missing todo -> 404	YES	404		session2_log.txt S2-14
/todos/:id/categories	GET	DOCUMENTED	missing todo -> 404	NO	200, returns category 1	#BUG	session2_log.txt S2-15
/todos/:id/categories	HEAD	DOCUMENTED	missing todo -> 404	NO	200	#BUG	session2_log.txt S2-16
/todos/:id/categories	POST	DOCUMENTED	link with XML body	NO	404 'could not find thing' even though category 1 exists	#BUG	session2_log.txt S2-17
/todos/:id/categories	POST	UNDOCUMENTED	title instead of id -> 400?	NO	201, made a new category 'Errands' and linked it	not in docs	session2_log.txt S2-18, S2-20
/todos/:id/categories/:id	DELETE	DOCUMENTED	unlink	YES	200		session2_log.txt S2-12
/todos/:id/categories/:id	DELETE	DOCUMENTED	unlink again -> 404	YES	404		session2_log.txt S2-13
/todos/:id/categories/:id	OPTIONS	UNDOCUMENTED	allowed methods	YES	200, Allow: OPTIONS, DELETE	not in docs	session2_log.txt S2-24
/todos/:id	PUT	DOCUMENTED	title only, links kept	NO	200 but category AND project links gone	#BUG same PUT issue as session 1	session2_log.txt S2-19 to S2-23